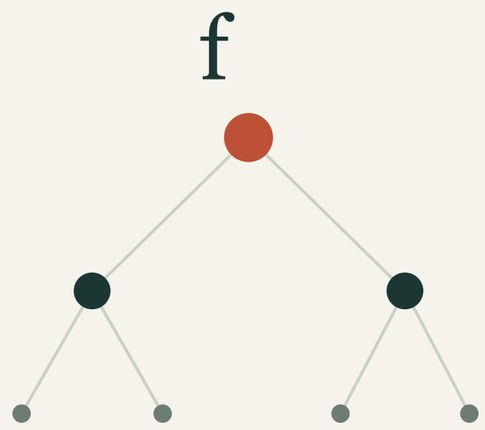
PAPER v1.5.8

Numbers, constructed.



FROM NATURAL NUMBERS TO GENERATIVE QUATERNIONS

One finite syntax. More numbers, with explicit rules at every step.

01 **N**

Natural numbers

$0 = f()$ $S(n) = f(n)$

$f() \to f(f()) \to f(f(f()))$

Base and successor.
Church encoding for λ.

02 **Z**

Signed integers

$(s, n), \quad n \in N$

s is positive or negative; zero has positive sign.

Pair a sign
with a magnitude.

03 **Q**

Rational numbers

$p / q, \quad p, q \in Z, \quad q > 0$

Reduce fractions; normalize zero to 0/1.

Integer ratios form
an ordered field.

04 **L**

Linear relation labels

$(k, a) \mapsto a + k\varepsilon$

$k \in Z$, $a \in Q$; $q\#\# = (2, q) \mapsto q + 2\varepsilon$

k counts relation operators,
not recursion steps.

05 **M**

M-polynomials

$P(\varepsilon) = a_0 + a_1\varepsilon + \cdots + a_n\varepsilon^n$

Finite rational coefficient list; trim trailing zeros.

Coefficient positions
encode powers of ε.

06 **F_ε**

Finite rational-expression numbers

$P(\varepsilon) / Q(\varepsilon), \quad Q \neq 0$

$P, Q \in M$; $P/Q = R/S \Leftrightarrow PS = RQ$

$0 < \varepsilon < r$
for every positive rational r.

07 **C_ε**

Generative complex numbers

$$J(a, b) = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$$

$a, b \in F\varepsilon$; $i = J(0, 1)$, $i^2 = -1$

Two coefficients,
one structured matrix.

08 **H_ε**

Generative quaternions

$(z, w) \mapsto z + wj$

$z = J(a, b)$, $w = J(c, d)$; $a, b, c, d \in F\varepsilon$

$j^2 = -1$
$ij = k, \ ji = -k$

COMMON SYNTAX

$$U := f() \mid f(U) \mid f(UU)$$

$(u, v) \mapsto f(E(u)E(v))$

Read at a specified type. Operations and equality have explicit, type-specific rules.